

Introduction to Image Processing

Assignment 2

Deadline: 9th June EOD

Instructions

- Attempt all questions.
 - Write clean and readable code.
 - Explain theoretical answers in your own words in the text cell in notebook (.ipynb) not as comments in code cell
 - Use comments wherever necessary.
 - You may use NumPy and Pillow library.
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Question 1: Understanding Digital Images (Easy)

A grayscale image is represented by:

$$\begin{bmatrix} 20 & 40 & 60 \\ 80 & 100 & 120 \\ 140 & 160 & 180 \end{bmatrix}$$

Part A (Theory)

Answer the following:

1. What is a pixel?
2. What does the value 0 represent in a grayscale image?
3. What does the value 255 represent?
4. How many pixels are present in the image above?

Part B (Coding)

Write a Python program that:

1. Stores the matrix using NumPy.
 2. Prints image dimensions.
 3. Finds:
 - Minimum pixel value
 - Maximum pixel value
 - Average pixel value
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Question 2: RGB Image Analysis (Medium)

Consider the RGB pixel values:

(255,0,0)
(0,255,0)
(0,0,255)
(255,255,0)
(255,255,255)

Part A (Theory)

Answer the following:

1. What does RGB stand for?
2. Why are three channels used?
3. What color is produced by:

- (255,255,0)
- (255,0,255)
- (0,255,255)

4. Explain why RGB is suitable for digital displays.

Part B (Coding)

Create a NumPy array containing at least 10 RGB pixels.

Write a program that:

1. Calculates average Red value.
2. Calculates average Green value.
3. Calculates average Blue value.
4. Finds the brightest pixel using:

$$Brightness = \frac{R + G + B}{3}$$

Question 3: Image Cropping and Resizing (Medium)

Load any image using Pillow. *(If you are reading this that means you are not using any LLM directly, so please add any funny **donald trump** image here :-)*

Part A (Theory)

Answer the following:

1. What is cropping?
2. What is ROI (Region of Interest) and it's use case?
3. What is aspect ratio, and why it should be preserved?

Part B (Coding)

Write a program that:

1. Loads an image.
2. Displays original dimensions.
3. Crops the center region.
4. Resizes the cropped image to:
 - 300×300
 - 600×600
5. Save all generated images.

Part C (Analysis)

Compare the original image and resized images.

Discuss:

- Quality changes
 - Distortion (if any)
 - Aspect ratio effects
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Question 4: Interpolation Investigation (Hard)

Different interpolation techniques produce different results.

Part A (Theory)

Research and explain:

1. Nearest Neighbor Interpolation
2. Bilinear Interpolation
3. Bicubic Interpolation
4. Lanczos Interpolation

For each method discuss:

- Working principle
- Advantages
- Disadvantages
- Suitable applications

Part B (Coding)

Using the same image:

1. Resize the image to 100x100.
2. Resize it back to 800x800 using:
 - Nearest Neighbor
 - Bilinear
 - Bicubic
 - Lanczos
3. Save all outputs.

Part C (Analysis)

Compare all outputs and answer:

1. Which method preserved the most detail?
 2. Which method appeared blurriest?
 3. Which method is best for pixel art?
 4. Which method would you choose for CubeCanvas and why?
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Question 5 (Mini Project - 50 Marks)

Image Processing Explorer

Create a Python application that demonstrates all major concepts learned in this module.

Your application should:

1. Load an image. (*Spongebob image pl ;-)*)
2. Display image dimensions.
3. Display aspect ratio.
4. Resize the cropped image to user-specified dimensions.
5. Generate outputs using:
 - Nearest Neighbor
 - Bilinear
 - Bicubic
 - Lanczos
6. Display pixel values at user-selected coordinates.
7. Compute:
 - Average Red value
 - Average Green value
 - Average Blue value
8. Save all generated images automatically.

Bonus Features (Optional)

Implement any two:

1. Brightness adjustment.
2. Contrast adjustment.
3. Grayscale conversion.
4. Image rotation.
5. Histogram visualization.